



OPEN SOURCE ENDOWMENT

Members call
19 May 2026

Agenda

1. Open Source Endowment (Chad Whitacre)
2. Project selection model (Konstantin Vinogradov)
3. Discussion and Q&A

OSE Overview



\$755K
endowment fund

110
total donors

67
members (\$1K+ donors)



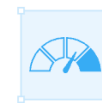
Sustainable

Provides a stable, long-term funding model, independent of volatile corporate and personal budgets.



Systemic

Supports the critical but underfunded open source projects that conventional models miss.



Efficient

A lean, digital-first, community-led organization designed for maximum impact and transparency.

OSE Team

BOARD OF DIRECTORS



Konstantin Vinogradov
Chairman

VC investor in open source, AI and infra software. Former General Partner at Runa Capital (\$600M+ AUM). Creator and maintainer of the ROSS Index (2020–2025), a quarterly data-driven index of open-source startups.



Chad Whitacre
Secretary

Head of Open Source at Sentry (used by 4M+ developers), who leads the Open Source Pledge, a cultural initiative to get companies to pay maintainers. Founded open source funding pioneer Gratipay (2012–2018).



Maxim Konovalov
Treasurer

Co-founder and former CTO at Nginx, the world's leading web server used by 400M+ websites, which was acquired for ~\$700M in 2019. Former VP Engineering at F5. Long-time contributor to FreeBSD.

STAFF & ADVISORS



Jonathan Starr
Executive Director

Former Program Manager at NumFOCUS, a nonprofit supporting scientific OSS such as Pandas. Co-Founder of SciOS and The Institute of Open Science Practices, focusing on sustainable open science.



Amy Parker
Advisor

Deputy Executive Director at the OpenSSL Foundation and former Director of the Wikipedia Endowment. Served as Director of Individual Giving at the Smithsonian and the New York Public Library.



Vlad-Stefan Harbuz
Advisor

Maintainer of the Open Source Pledge. Core developer of OSS funding service thanks.dev. Helped build software used by the Gates Foundation to allocate \$1B in healthcare funding.



Selection Model

(draft for discussion)

Model Approach

Targets

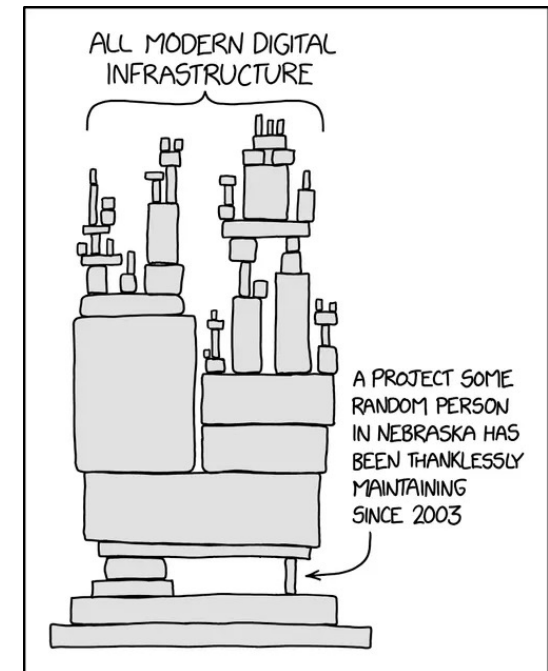
- Most risky out of the most valuable projects
- Independent OSS without corporates or startups behind them

Model

- Open, transparent, measurable, auditable
- Algorithmic: data-driven + human-in-the-loop

Format

- Microgrants up to \$5000 (a la MacArthur Fellowship)
- Project leaders distribute funds internally



Model Pipeline

VALUE 	RISK 	DUE DILIGENCE 
• Downloads • Dependencies • Dependents	• Concentration • Complexity • Security • Maintainer workload • Funding	• Eligibility: OSS, etc. • Data and intent validation • KYC/AML and other legal checks
Defines “core” projects (power law distribution)	Finds the riskiest projects within the target scope	Manual checks top candidates before funding them

Design rule: with all other factors taken equal, a component's growth should clearly increase/decrease Value or Risk, but not both

Value Data

Area	Sources
Python	 
JavaScript / TypeScript	
Rust	
C / C++	  
Repo data	

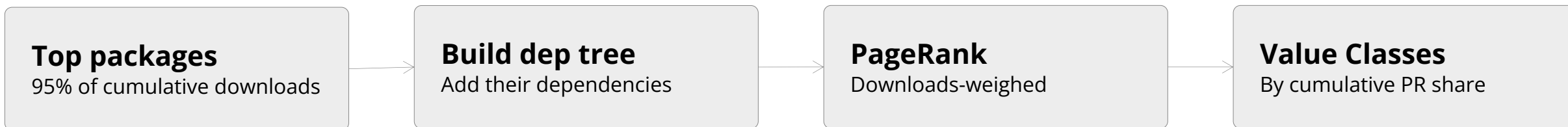
Current limitations

- Only GitHub for now, full git support in the future
- Lack of verified package-repo links (PyPI, C/C++)
- Approximated annual downloads for Homebrew
- Only runtime dependencies for C/C++
- No connections between ecosystems yet

Current draft

- <https://github.com/osendowment/model>

Value PageRank



Key ideas

- Based on annual data for 2021-2025
- (Downloads, dep graph) → PageRank
- Focus on classes, not specific numbers
- Classes by cumulative PR: A (0-50%), B(50-75%), C (75-90%)

Output:

- Sorted list of git urls and github repos with classes (ABCD)

Eco	Packages	Git Repos	A	B	C	D	%GH AB
npm	6,370	4,413	143	428	765	3,077	100%
pypi	3,139	2,526	42	121	306	2,057	97%
crates	6,218	3,576	31	101	253	3,191	100%
cpp	1,882	1,860	10	80	289	1,481	78%
Total	17,609	12,333	226	728	1,610	9,769	97%

Repos accounting for 95% of downloads on npm, PyPI, Crates, Debian popcon, Homebrew

Value Python

Independent Python packages (class A)

Package	Org	Avg dL	PR
six	benjaminp	3.4B	0.0196
pycparser	eliben	2.1B	0.0178
cffi	python-cffi	2.3B	0.0174
certifi	certifi	4.3B	0.0116
idna	kjd	4.2B	0.0114
colorama	tartley	1.8B	0.0104
pyasn1	pyasn1	2.0B	0.0096
charset-normalizer	jawah	4.0B	0.0093
tomli	hukkin	1.2B	0.0071
brotlicffi	python-hyper	5M	0.0061
zipp	jaraco	2.0B	0.0059
ruff	astral-sh	230M	0.0058
fsspec	fsspec	2.2B	0.0043
pyasn1-modules	pyasn1	1.3B	0.0043
pygments	pygments	1.3B	0.0042
rsa	sybrenstuvcl	2.0B	0.0035

Umbrella-org Python packages (class A)

Package	Org	Avg dL	PR
typing-extensions	python	4.1B	0.0496
urllib3	urllib3	5.5B	0.0189
python-dateutil	dateutil	3.6B	0.0148
boto3	boto	9.5B	0.0141
packaging	pypa	3.3B	0.0136
numpy	numpy	2.8B	0.0109
cryptography	pyca	2.6B	0.0096
s3transfer	boto	3.5B	0.0094
markupsafe	pallets	2.0B	0.0090
importlib-metadata	python	2.2B	0.0087
pytest	pytest-dev	1.3B	0.0086
setuptools	pypa	4.4B	0.0080
brotli	google	134M	0.0063
pyyaml	yaml	3.0B	0.0061
google-auth	googleapis	1.5B	0.0056
aiobotocore	aio-lib	3.0B	0.0052
tzdata	python	981M	0.0050
click	pallets	2.2B	0.0050
aiohttp	aio-lib	1.4B	0.0041
wheel	pypa	2.5B	0.0041
pip	pypa	2.6B	0.0037
pandas	pandas-dev	2.2B	0.0037
pydantic	pydantic	1.8B	0.0037
jinja2	pallets	1.9B	0.0036

Value C/C++

Projects with formal organizational umbrella (FSF, consortium, foundation):

Rank	Class	Package	PR	Score	Organization
1	A	glibc	0.2147	674,803	GNU / FSF
2	A	gcc	0.0575	1,428,966	GNU / FSF
8	A	gettext	0.0192	2,500,621	GNU / FSF
11	A	gmp	0.0144	1,674,133	GNU / FSF
14	B	openssl	0.0088	3,817,246	OpenSSL Software Foundation
17	B	mpfr	0.0071	856,697	INRIA + GNU / FSF
21	B	libunistring	0.0077	1,646,601	GNU / FSF
18	B	readline	0.0044	2,814,435	GNU / FSF
25	B	ncurses	0.0037	1,284,533	GNU / FSF (Thomas Dickey lead)
26	B	sqlite	0.0039	4,010,976	SQLite Consortium
31	B	freetype	0.0026	1,821,308	FreeType Project (nonprofit)
34	B	icu	0.0025	1,644,083	Unicode Consortium
40	B	gnutls	0.0021	1,818,557	GNU / FSF
42	B	fontconfig	0.0021	1,906,280	freedesktop.org
46	B	libidn2	0.0020	1,945,724	GNU / FSF
50	B	libpgp-error	0.0019	921,836	GnuPG e.V.
52	B	krb5	0.0018	1,374,768	MIT Kerberos Consortium
54	B	libxml2	0.0017	837,979	GNOME Foundation

Projects without formal organizational backing – single-person or small-volunteer teams:

Rank	Class	Package	PR	Score	Maintainer(s)	Bus factor
9	A	zlib	0.0159	1,210,201	Mark Adler, Jean-loup Gailly	2
12	B	xz / liblzma	0.0095	4,172,412	Lasse Collin (Tukaani)	1
24	B	pcres	0.0030	2,306,580	Philip Hazel	1
12-B	B	Boehm-gc	0.0055	659,948	Hans-J. Boehm	1
35	B	nghttp2	0.0024	2,810,746	Tatsuhiro Tsujikawa	1
48	B	bzip2	0.0020	703,522	Julian Seward (dormant)	0-1
49	B	lcms	0.0019	1,688,891	Marti Maria (LittleCMS)	1
53	B	expat	0.0017	744,506	libexpat community	~3
57	B	curl	0.0016	1,341,095	Daniel Stenberg (wolfSSL sponsored)	1
67	B	c-ares	0.0014	1,294,769	c-ares maintainers	~2
68	B	giflib	0.0014	1,117,583	giflib maintainers	~2
71	B	libffi	0.0013	809,465	libffi project	~3
73	B	pcres2	0.0013	278,606	Philip Hazel	1
78	B	ffmpeg	0.0013	1,335,595	FFmpeg volunteer collective	~5
80	B	libyaml	0.0012	754,482	libyaml maintainers	~2

Risk Data – work in progress

Target scope: < 1000 top repos responsible for 75% of cumulative PageRank (“value AB classes”)

Concentration	Complexity	Security	Workload	Funding
• Bus Factor (BF) • Herfindahl-Hirschman Index (HHI)	• LOCs • Cyclomatic complexity (CC)	• OpenSSF score • CVEs	• Active maintainers (AM) • LOC / AM • CVE / AM • Net new issues / AM	• GitHub Sponsors • Funding.json
 	 	 	 	 

Risk Concentration & Complexity

Repository ^	> Results					Concentration					Complexity				
	Risk P ^	BF ^	HHI ^	P ^	C ^	CC ▼	LOC ^	SLOC ^	P ^	C ^					
gcc-mirror/gcc	22	183	56	1	D	1,469,572	13.4M	9.3M	100	A					
nodejs/node	25	19	200	3	D	828,445	14.6M	11.8M	100	A					
azure/azure-sdk-for-python	37	12	434	6	D	601,121	12.4M	9.8M	100	A					
swc-project/swc	46	2	2,696	31	C	537,832	4.4M	3.6M	99	A					
rust-lang/rust	26	27	291	3	D	231,548	3.6M	2.7M	99	A					
ffmpeg/ffmpeg	35	8	731	10	D	222,538	2M	1.5M	99	A					
unicode-org/icu	24	6	672	10	D	203,491	2M	1.4M	99	A					
vercel/next.js	37	12	327	5	D	162,265	2M	1.8M	99	A					
openssl/openssl	29	6	630	10	D	150,022	1.2M	928.9K	98	A					
azure/azure-sdk-for-js	39	10	464	7	D	149,057	7M	5M	99	A					
systemd/systemd	31	3	1,286	19	D	145,753	1M	767.3K	98	A					
microsoft/typescript	24	8	410	7	D	144,381	4.1M	2.5M	99	A					
microsoft/windows-rs	66	1	4,953	61	C	101,065	2.3M	2.3M	99	A					
protocolbuffers/protobuf	41	8	528	8	D	85,482	1.1M	820.4K	98	A					
grpc/grpc	40	9	549	8	D	78,931	1.2M	953.7K	98	A					

[swc-project](#) and [systemd/systemd](#) are examples of high-risk repos:

- very complex (millions of LOC)
- very concentrated around a few contributors (bus factors of 2–3)

Risk Security

	▼ Results				> Concentration	> Complexity	▼ Security			
Repository ▲	Value ▲	Value P ▲	Risk ▲	Risk P ▲	P ▲	P ▲	OpenSSF ▲	CVEs ▼	P ▲	Active
ffmpeg/ffmpeg	B	26	A	35	10	99	5.1	150	64	
apache/airflow	B	43	A	26	4	98	7.8	90	18	
curl/curl	B	29	B	29	50	95	8.9	88	7	
openssl/openssl	B	48	A	29	10	98	7.1	73	31	
nodejs/node	B	33	A	25	3	100	6.1	59	50	
python-pillow/pillow	B	35	B	33	28	91	8.6	46	9	
gnome/libxml2	B	30	B	40	5	94	4.2	45	74	
git/git	B	29	B	50	14	97	5.3	40	61	
libexpat/libexpat	B	30	C	42	33	82	8.0	40	15	
lnetlabs/unbound	B	29	B	58	69	94	5.9	34	52	
vercel/next.js	B	26	A	37	5	99	6.0	27	51	
langchain-ai/langchain	B	28	B	29	3	94	7.2	25	29	
systemd/systemd	B	49	A	31	19	98	8.9	23	7	
webmproject/libwebp	B	30	B	65	33	90	3.9	20	82	
c-ares/c-ares	B	28	C	53	26	86	6.3	18	47	
krb5/krb5	B	30	B	44	9	96	5.3	18	61	
pyca/cryptography	A	64	B	38	26	88	8.4	16	13	
aio-lib/aiohttp	B	48	C	36	16	87	7.4	15	26	

Does a high number of CVEs indicate low security? Probably not in the case of [curl/curl](#)

It is crucial to combine a few metrics within one domain — for instance, CVEs and the OpenSSF score for security

Risk Funding

	> Results	> Concentration	> Complexity	> Security	> Workload	~ Funding	
Repository ▲	Risk P ▲	P ▲	P ▲	P ▲	P ▲	GH Sponsors ▲	P ▲
servo/rust-fnv 	7	14	20	49	—	271	0
servo/rust-smallvec 	8	6	57	71	—	271	0
servo/rust-url 	9	20	72	29	—	271	0
vitest-dev/vitest 	14	24	91	48	—	200	0
sindresorhus/find-cache-directory	10	32	14	44	—	192	1
sindresorhus/package-directory	10	66	5	45	—	192	1
sindresorhus/read-package-up 	10	70	6	46	—	192	1
sindresorhus/ansi-escapes 	25	66	25	49	—	184	5
sindresorhus/array-union 	11	60	1	54	—	184	5
sindresorhus/binary-extensions 	10	51	1	52	—	184	5
sindresorhus/callsites 	16	63	4	54	—	184	5
sindresorhus/camelcase 	26	55	33	49	—	184	5
sindresorhus/clean-stack 	25	63	25	52	—	184	5
sindresorhus/cli-cursor 	21	70	10	54	—	184	5
sindresorhus/crypto-random-string 	25	55	26	52	—	184	5
sindresorhus/decamelize 	22	58	16	52	—	184	5
sindresorhus/detect-newline 	19	66	7	54	—	184	5
sindresorhus/environment 	22	81	10	52	—	184	5
sindresorhus/escape-string-regexp 	15	50	4	52	—	184	5

Some developers are skilled at attracting sponsors, but most of them are not and are unlikely to ever be.

Lack of funding is one of the key risk components, which can only partially be assessed via automated data checks.

Due Diligence

ELIGIBILITY 	VALIDATION 	KYC/AML 
• Open Source license • Not an end-of-life project • Independent - no corporate trademarks - no associated startups - community-led	Candidates fill in a small application: • Key people, leader, country • Funding, governance • Payment details, legal checks, etc	Depending on the countries and other details, OSE invites grantees to receive funding via relevant services like: • GitHub Sponsors • Stripe Connect
OSE contacts the top eligible candidates	OSE validates key declared facts. The board approves the top projects	OSE relies on KYC/AML of notable US payment platforms

**Let's discuss this
model and any other topics**

Questions

Quick asks

1. Nominate projects at <https://endowment.dev/funding>
2. Do we need a member-only group? Email/WhatsApp/etc?

Open questions

- The best way to combine Risk metrics?
- Any low-hanging fruit among data sources left?
- C/C++ ecosystem — contributors wanted
- Any other questions 😊